

CS 458

Spring 2025: Assignment 4

Due in Canvas Assignment Tuesday, April 22, 2025 by 11:59 pm

Task1: True/False Questions (3.5 points)

1. **True:** In symmetric cryptography, the same key is used for both encryption and decryption.
2. **True:** A public-key cryptosystem eliminates the need for secure key distribution.
3. **False:** Public-key cryptography uses two keys: a public key and a symmetric key.
4. **True:** Digital signatures use a private key to sign a message and the public key to verify the signature.
5. **True:** A trapdoor one-way function is hard to reverse unless you have specific secret information.
6. **False:** Asymmetric cryptography is generally faster than symmetric cryptography.
7. **False:** RSA relies on the difficulty of solving discrete logarithms.
8. **False:** A key size of 128 bits provides the same level of security for both symmetric and asymmetric systems.
9. **True:** The square-and-multiply algorithm optimizes modular exponentiation in RSA.
10. **False:** RSA without padding is semantically secure and resistant to chosen ciphertext attacks.
11. **True:** The OAEP padding scheme is used to improve RSA security.
12. **True:** Diffie-Hellman key exchange is vulnerable to man-in-the-middle attacks if not authenticated.
13. **True:** The security of RSA depends on the factoring problem being computationally hard.
14. **True:** In Diffie-Hellman, both parties end up with the same shared secret key.
15. **False:** Quantum computers have no known impact on public key cryptography like RSA.

Task2: Multiple Choice Questions (MCQs) (3.5 points)

1. Which of the following is a problem with symmetric cryptography?
 - A. Slow performance
 - B. Insecure algorithms
 - C. Key distribution**
 - D. No confidentiality
2. Which property does public-key cryptography ensure using digital signatures?
 - A. Confidentiality
 - B. Non-repudiation**
 - C. Speed
 - D. Multiplexing
3. Which is NOT a use of public-key cryptography?
 - A. Key distribution
 - B. Bulk data encryption**
 - C. Digital signatures
 - D. Confidentiality
4. RSA is based on which mathematical problem?
 - A. Discrete logarithms
 - B. Factoring large integers**
 - C. Solving polynomials
 - D. Graph isomorphism
5. What is Euler's Totient Function $\phi(N)$ used for in RSA?
 - A. Calculating ciphertext
 - B. Choosing the public key
 - C. Generating key pairs**
 - D. Determining primality
6. In RSA, what is the result of decrypting ciphertext $C = M^e \bmod N$?
 - A. A hash of the message
 - B. The public key
 - C. The original message M**
 - D. The product of two primes
7. What is the main advantage of square-and-multiply over naïve modular exponentiation?
 - A. Requires no exponent
 - B. Faster and more efficient**
 - C. Stronger encryption
 - D. Better for decryption only
8. Which of the following is a recommended RSA public exponent for efficiency and security?
 - A. 7
 - B. 17
 - C. 65537**
 - D. 2048
9. Which property of RSA allows attackers to modify encrypted messages?
 - A. Determinism
 - B. Asymmetry

C. Malleability

D. One-wayness

10. What is the main purpose of padding schemes in RSA?

A. Make encryption faster

B. Prevent key reuse

C. Ensure semantic security

D. Increase modulus size

11. In OAEP, which operation is performed first when decrypting?

A. Encrypting message

B. Recovering random string r

C. Hashing the message

D. Padding removal

12. Which key size is recommended by NIST for RSA until 2030?

A. 1024 bits

B. 2048 bits

C. 3072 bits

D. 512 bits

13. What does the Diffie-Hellman key exchange securely produce?

A. A public key

B. A shared symmetric key

C. A digital signature

D. A message digest

14. Which attack does not target the mathematical structure of RSA but its implementation?

A. Brute force

B. Timing attack

C. Factoring attack

D. Chosen ciphertext attack

15. Which future technology poses a serious threat to RSA security?

A. 5G

B. Blockchain

C. Quantum computing

D. GPUs

Task3: Break The RSA (3 points):

We have an RSA public-key cryptosystem with a public key ($N=39$, $e=5$).

1. (2 points) Break the system and find: P , q , $\phi(N)$, private key: d .

(Show your work in detail)

2. (1 points) We have a cipher-text: $C=3$. Recover corresponding plain-text M from this cipher-text using your calculated private key. (Show your work)

All of my work is in the picture below:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29

RSA HW4
 $N=39$, $e=5$
 Find p , q , $\phi(N)$, and d

$N = p \times q = 39$
 3×13
 $p=3$ $q=13$

$\phi(N) = \phi(p) \times \phi(q) = (p-1) \times (q-1) = 2 \times 12 = 24$
 $\phi(N) = 24$

d is in range $\in [1, \phi(N)-1] = \in [1, 23]$

d must satisfy $e \times d \bmod \phi(N) = 1$
 $5 \times d \bmod 24 = 1$
 $d = 5$

$C = 3$
 $M = C^d \bmod N$
 $= 3^5 \bmod 39$
 $= 243 \bmod 39$

$243 \div 39 = 6 \text{ Remainder } 9$

$M = 9$